

LAB 8: DEADLOCK DETECTION ALGORITHM

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C PROGRAM

```
#include <stdio.h>
int main()
{
    int allocation[10][10], request[10][10];
    int available[10];
    int work[10];
    int finish[10];
    int n, m;
    int i, j, k;
    int found;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);
    printf("Enter Number of Resource Types: ");
    scanf("%d", &m);

    printf("
Enter Allocation Matrix:
");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("
Enter Request Matrix:
");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            scanf("%d", &request[i][j]);
        }
    }

    printf("
Enter Available Resources:
");
    for(i = 0; i < m; i++)
    {
        scanf("%d", &available[i]);
        work[i] = available[i];
    }

    for(i = 0; i < n; i++)
    {
        finish[i] = 0;
    }
}
```

```

do
{
    found = 0;
    for(i = 0; i < n; i++)
    {
        if(finish[i] == 0)
        {
            for(j = 0; j < m; j++)
            {
                if(request[i][j] > work[j])
                    break;
            }
            if(j == m)
            {
                for(k = 0; k < m; k++)
                {
                    work[k] += allocation[i][k];
                }
                finish[i] = 1;
                found = 1;
            }
        }
    }
} while(found);

found = 0;
printf("
Deadlocked Processes:
");
for(i = 0; i < n; i++)
{
    if(finish[i] == 0)
    {
        printf("P%d ", i);
        found = 1;
    }
}

if(found == 0)
{
    printf("No Deadlock Detected");
}
printf("
");
return 0;
}

```

SHELL SCRIPTING

```

#!/bin/bash
echo "Enter number of processes:"
read n
echo "Enter deadlocked process numbers (if any):"
read processes
if [ -z "$processes" ]
then
    echo "No Deadlock Detected"
else
    echo "Deadlocked Processes:"

```

```
    echo "$processes"  
fi
```

OUTPUT

```
(kali@Harish)-[~]  
└─$ nano deadlock.sh  
(kali@Harish)-[~]  
└─$ chmod +x deadlock.sh  
(kali@Harish)-[~]  
└─$ bash deadlock.sh  
Enter number of processes:  
2  
Enter deadlocked process numbers (if any):  
1  
Deadlocked Processes:  
1
```